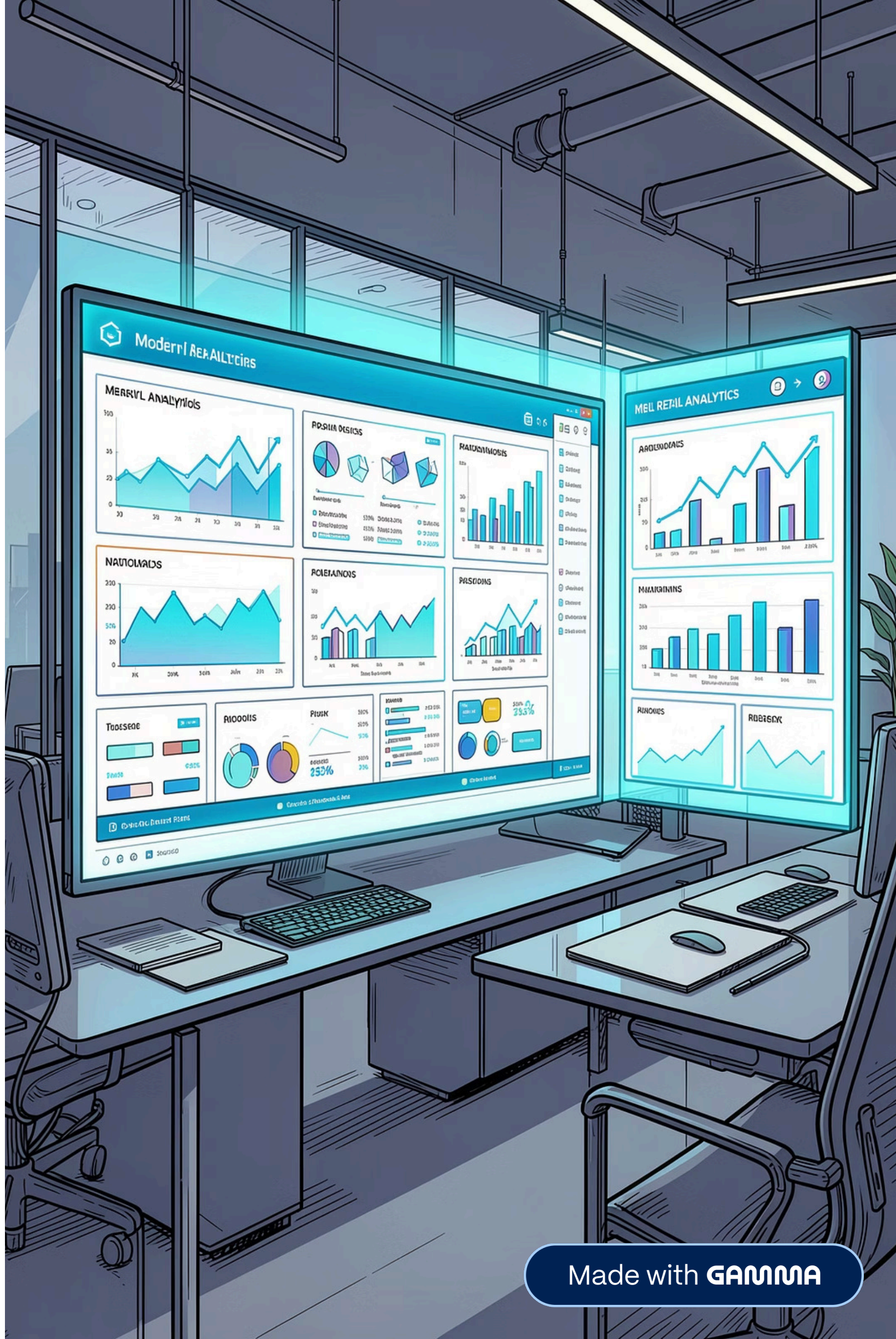
Customer Shopping Behaviour Analysis

Uncovering spending patterns, customer segments, and product preferences from 3,900 transactions — to guide smarter retail strategy.

DATA SCIENCE · SQL · POWER BI



Project at a Glance

This project analyses transactional retail data to surface actionable insights across customer demographics, purchase behaviour, and subscription trends.

3,900

Purchases Analysed

Transactional records across multiple product categories

18

Data Columns

Demographics, purchase details, and behavioural signals

37

Missing Values

Imputed in Review Rating using category medians

5

Key Recommendations

Strategic actions derived from the analysis

Dataset Overview

Three Feature Groups

Customer Demographics

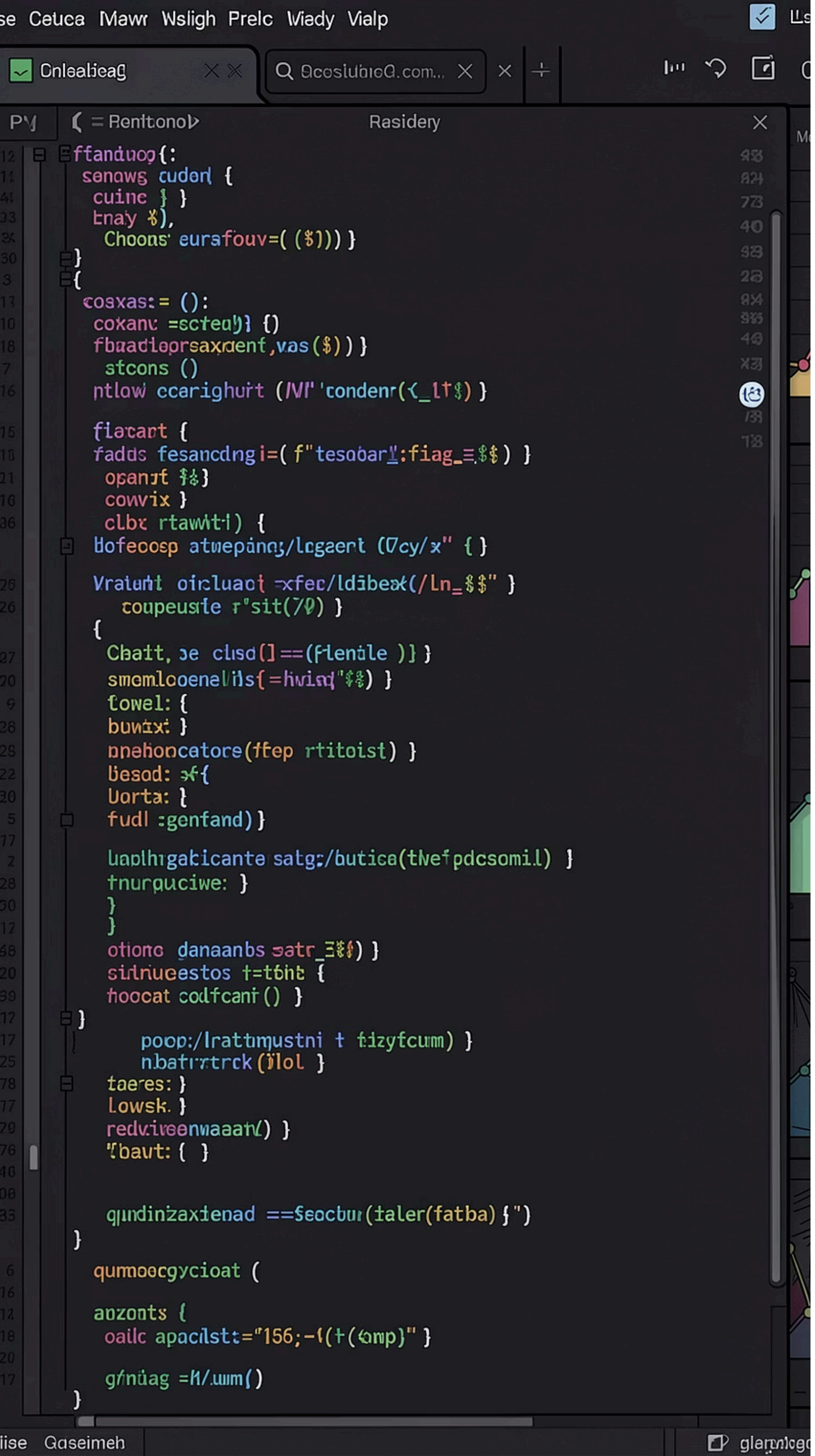
Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Colour

Shopping Behaviour

Discount, Previous Purchases, Frequency, Review Rating, Shipping Type



Data Preparation in Python

1

Load & Explore

Imported via pandas; inspected with `df.info()` and `.describe()`

2

Clean & Standardise

Imputed missing Review Ratings using category medians; renamed columns to snake_case

3

Engineer Features

Created `age_group` bins and `purchase_frequency_days` from raw dates

4

Load to Database

Removed redundant `promo_code_used`; exported cleaned DataFrame to MySQL

SQL Analysis: Revenue & Spending

Structured MySQL queries answered core business questions on revenue distribution and customer spend.

What We Investigated

- Gender-based revenue split to assess demographic contribution
- Whether discount users still outspend the average
- Which age groups drive the most revenue
- If Express shipping customers spend more than Standard

Subscribers vs. Non-Subscribers

Key Question

Do subscribers generate higher average spend and total revenue than non-subscribers?

We also examined whether customers with more than 5 purchases are more likely to hold a subscription — testing the loyalty-to-subscription hypothesis.

> Repeat buyers and subscription uptake are closely linked — a key lever for retention strategy.

Products, Ratings & Discounts

Top 5 Products by Rating

Identified highest-rated products by average review score to inform featured listings and campaigns.

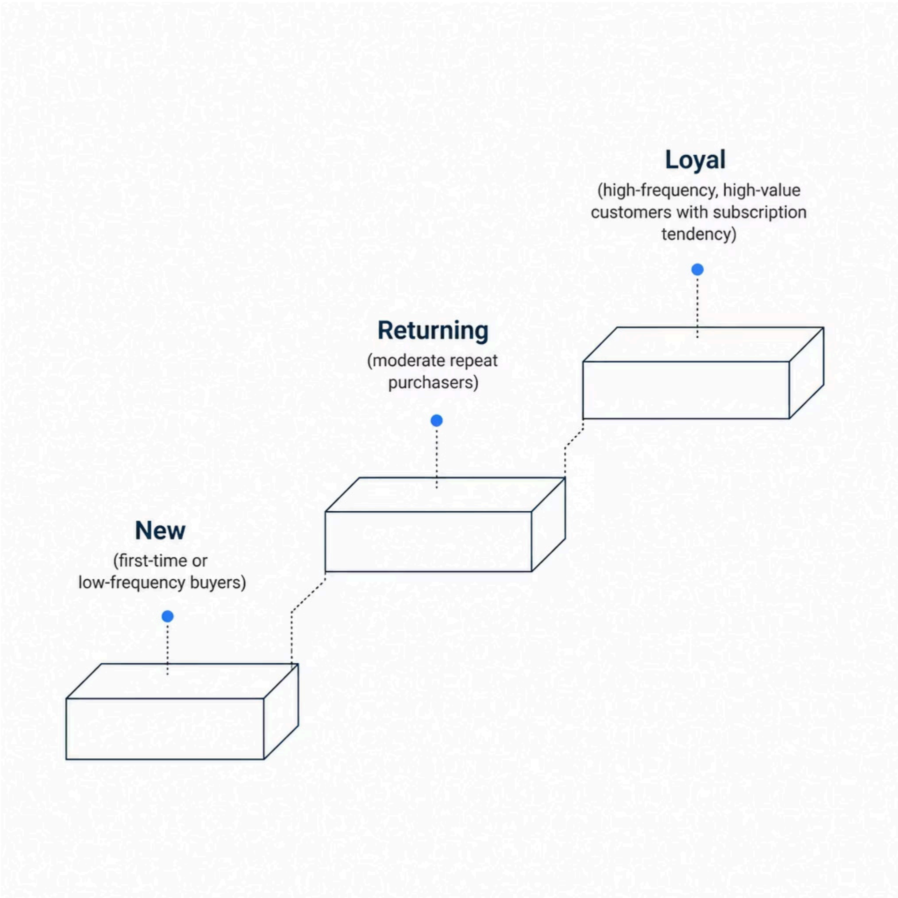
Discount-Dependent Products

Flagged the 5 products with the highest percentage of discounted purchases — signalling margin risk or price sensitivity.

Top 3 Products per Category

Ranked best-selling items within each category to guide inventory and promotional focus.

Customer Segmentation



Segment Logic

Customers were classified into three tiers based on purchase history and frequency:

New

First purchase or very low frequency

Returning

Moderate repeat purchases, growing engagement

Loyal

High frequency, high value — prime subscription candidates

Power BI Dashboard

A fully interactive dashboard was built to translate SQL findings into visual, stakeholder-ready insights — enabling self-service exploration of revenue, segments, and product performance.

Revenue Views

By gender, age group, and shipping type

Segment Breakdown

New, Returning, and Loyal customer distribution

Product Performance

Top-rated and best-selling by category

Subscription Insights

Spend and uptake by subscriber status

Business Recommendations

→ **Boost Subscriptions**

Promote exclusive subscriber benefits to convert high-frequency buyers.

→ **Launch Loyalty Programmes**

Reward repeat purchasers to accelerate progression to the Loyal segment.

→ **Review Discount Policy**

Balance sales uplift against margin erosion on discount-dependent products.

→ **Strengthen Product Positioning**

Feature top-rated and best-selling products prominently in campaigns.

→ **Target High-Value Segments**

Focus marketing on high-revenue age groups and Express shipping users.